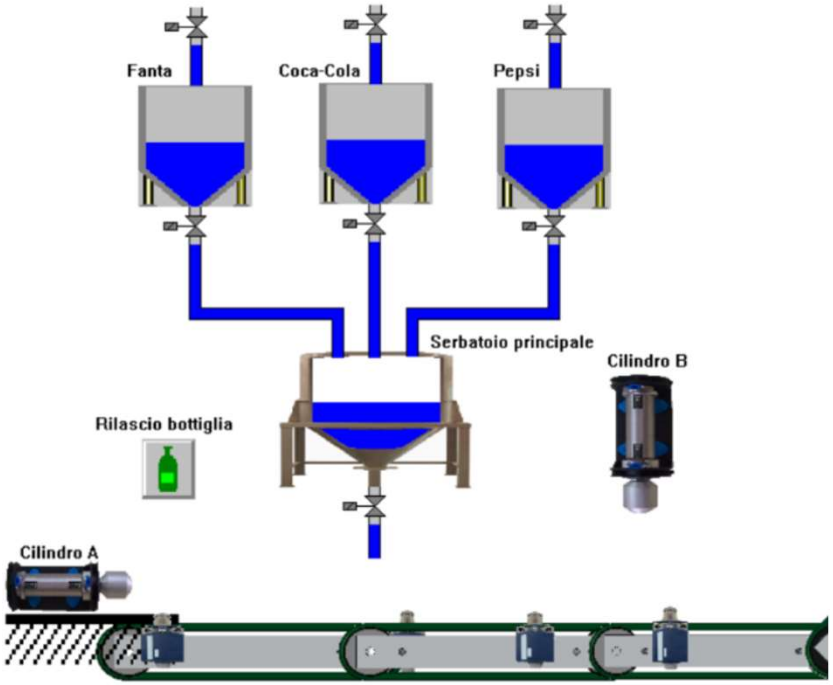
System Overview & Requirements



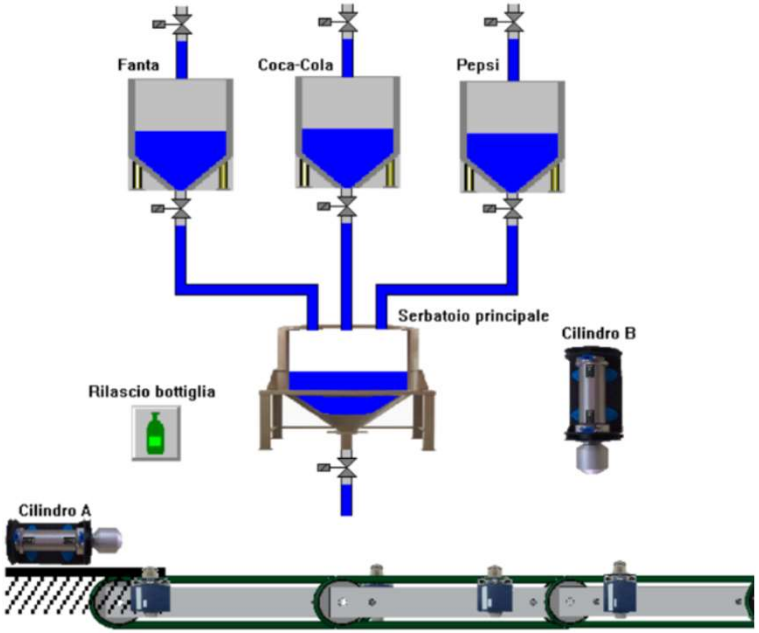
Bottle filling and sorting system.

Bottles of 3 typologies have to be filled and sorted (Fanta, Coca-cola and Pepsi as an example).
Figure above shows a possible interface of the application.

By means of proper selectors (*) the operator can choose the bottle to be filled down the filling tank (cylinder A pushes the bottles and a conveyor moves them; conveyor must be stopped during the filling).

Cylinder B simulates bottle corking (tapping); conveyor must be stopped during corking.

System Overview & Requirements



Define a tank at the end of the conveyor to collect a predefined numeber of bottles of the same type.
When the tank is full send the tank to a sorting system (only in the HMI: conveyor for Fanta, conveyoer for Coca-cola, conveyor for Pepsi; a pneumatic cylinder will push the tank on the conveyor to start the motors).

Do not forget to fill the «beverage tank» as soon as it is becoming empty.

Assignment

Develop the automatic control to implement the task using Codesys programming (for the logic) and FluidSim (plant and automation).

In particular develop a well organized code that uses both POU in LD, ST and FBD (batch method).

Organize the programming code so to include FB (Function block) with communication I/O signals and Fuction (FUN):

- Main POU in LD (Start, Stop, general commands);
- FBs with I/O communications for: cycle management (ST), actions execution (LD), Emergency (LD), HMI visibility (ST);
- FBs in FBD language (Function block diagram) for cylinders and objects movement/animation.

Moreover, develop a clear and understendable HMI interface in Codesys with: Start, Stop, Emergency, alarms, conveyor, tanks and bottles filling and emptying, cylinders and system animation … and all sensors and pilot lights needed to simulate the real plant.